Ollscoil Mhá Nuad
Maynooth University

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SEMESTER 1
2025-2026

CS416
Cryptography

Dr. T. De Wille, Dr. A. Mooney, Dr. T. Dowling

Time allowed: 2 hours

Answer all questions.

Instructions

	Yes	No	N/A
Formulae and Tables book allowed *(i.e. available on request)*		X	
Formulae and Tables book required *(i.e. distributed prior to exam commencing)*	X		
Statistics Tables and Formulae allowed *(i.e. available on request)*		X	
Statistics Tables and Formulae required *(i.e. distributed prior to exam commencing)*		X	
Dictionary allowed *(supplied by the student)*		X	
Non-programmable calculator allowed	X		
Students required to write in and return the exam question paper		X	

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[40 marks]

1 (a) Consider the number. [14 marks]

P= 74235903

P contains a HEX subliminal message.
Using n=47 as the base, extract the decimal coefficients of the polynomial generated by n that P represents.
Construct the HEX message by converting these coefficients to base 16.

(b) Given the following RSA public key data, [13 marks]

n	e	p
125563	**7**	**409**

Compute

i) The Carmichael Function, L(n)
ii) The corresponding private key, d.
iii) The two hex value Key that was encrypted using this system to give C= 40957.

Note:

i) $C^{157} \bmod n = 6992$
ii) $C^{68} \bmod n = 1281$
iii) $C^{18} \bmod n = 24010$

(c) A basic LWE encryption produces the ciphertext [13 marks]

$$\left(\begin{pmatrix}\mathbf{76}\\ \mathbf{81}\\ \mathbf{68}\end{pmatrix}, \mathbf{44}\right)$$

Given that q =83 and the private key, s, is

$$\begin{pmatrix}\mathbf{82}\\ \mathbf{1}\\ \mathbf{0}\end{pmatrix}$$

i) Briefly outline the idea behind LWE.
ii) Recover the plaintext. Show your calculations clearly.

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[30 marks]

2 (a) Show that [10 marks]

$$(83)^{-1} \bmod (113) = 80$$

Where all values are in Hex. What part of AES uses this type of calculation?

(b) "AES in counter mode is completely safe." [20 marks]

i) Describe, with the aid of a suitable diagram the counter mode of operation.

ii) Describe, using suitable notation, how you can exploit a known plaintext attack against this mode to insert changed ciphertext into the original ciphertext without being noticed.

iii) Given the plaintext ciphertext pair

P (HEX)	C(HEX)
BAD	743

Recover the key.
Given plaintext **P '= FFF** construct the ciphertext **C'** that can replace **C** without detection.

iv) Write down the main components of the CIA triad. Which component of this triad was exploited by the above attack? Briefly suggest a fix.

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[30 marks]

3 (a) An employee has been fired from a company but left without disclosing their key, k. Given the partial keys recovered from the CEO and the CTO, [10 marks]

	CTO	CEO
p	23099	17191
k mod p	6934	4338

Recover the key.

(b) Given the following El Gamal Signatures (a, b, H(m)) [14 marks]

a	b	H(m)
144904	77065	32501
38923	46596	78487
38923	72019	71432

and the public key,

(p,g,y) = (160087, 141700,2915)

find the secret key, x.

Notes:

i) $134663^{-1} \bmod p-1 = 100637$

ii) $38923^{-1} \bmod p-1 = 121285$

(c) What is the purpose of Salting a password? [6 marks]
Show, using an appropriate diagram, how Keccak achieves salting. Is Keccak a good choice for salting passwords? Briefly justify your answer.

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DATA SHEET

ASCII values

Character	A-Z	a-z	0-9
Ascii value	65-90	97-122	48-57

Lucas sequence

$$L_0 = 2, L_1 = 1,$$
$$L_i = L_{i-1} + L_{i-2} \; i > 2$$

DSA signature scheme

$r = (g^k \bmod p) \bmod q$
$s=(h+xr)k^{-1} \bmod q$
$a=hs^{-1}(\bmod q)$
$b=rs^{-1}(\bmod q)$
$v = (g^a y^b \; (\bmod p)) \bmod q)$

QR formula

$a^{(p-1)/2} \bmod p$

Blum Prime Square Root

$a^{(p+1)/4} \bmod p$

El-Gamal Signature Scheme (a, b)

$(g^k \bmod p, (h-xa)k^{-1} \bmod(p-1))$

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CRT

Given $x = a_1 \bmod m_1$

$x = a_2 \bmod m_2$

$x = a_3 \bmod m_3$

.....

$x = a_n \bmod m_n$

Then $x = \sum_{i=1}^{n} a_i M_i N_i \bmod (M)$

where $M = m_1 m_2 \ldots\ldots m_n$

$$M_i = \frac{M}{m_i}$$

and $N_i = M_i^{-1} \bmod m_i$

Garner's Formula

For $x < pq$

If $x \equiv a \pmod p$ and $x \equiv b \pmod q$

$T = p^{-1} \bmod (q)$

$u = (b - a)T \bmod q$

$x = a + up$

Basic LWE

Encryption

$c_{aux} = A^T . r + e_{aux}$

$c_{msg} = t^T . r + e_{msg} + m'$

Decryption

$m = dec(c_{msg} - s^T . c_{aux})$